SAMPLING-UNIT AND COMPOSITION SENSITIVITY OF AN AUDIO DEEPFAKE LEADERBOARD

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ABSTRACT

Leaderboard comparisons often treat correlated evaluation trials as independent. On ASVspoof 2021 DF, 95% all-28-pair max-t bands for ΔEER exclude zero for five of the six pairs among the four organizer baselines when trials are resampled independently, but for none when speakers and spoofing attacks are resampled independently. A single-source SpoofCeleb analysis specified before score access changes the corresponding count from 6/6 to 3/6; a disclosed post-result decomposition gives 5/6 under speaker-only and 3/6 under attack-only perturbation. Across 12 registered score-blind composition policies, 6/12 within-cohort orderings reverse but 0/16 between-cohort orderings do; a disclosed outcome-informed search finds reversal witnesses as small as $r_{TV} = .010218$. Under an imposed low-EER random-effects DGP, all three candidate intervals fall below .90 coverage together in 11/24 cells, rejecting the prespecified universal-coverage criterion under that DGP. Across the tested policies and deletions, the stable fixed-score conclusion is separation between the four SSL detectors and four organizer baselines, not their within-cohort ranking. These are evaluation-design sensitivity results, not population confidence claims.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task's systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The 2026 edition adds a calibration study instead; calibration establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [6], and 2021 tested every pair [1]; both use [7], and the practice then lapsed. Both variants of the cited test [7] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Across the eight detectors, the one-way random-intercept ICC $\sigma_s^2/(\sigma_s^2 + \sigma_e^2)$ on bona-fide scores ranges from 0.15 to 0.77. ICC alone does not establish dependence in paired EER differences; that is measured directly below by matched perturbations of the joint scores and then checked externally at family level on ASVspoof 5.

We audit two evaluation-design choices on fixed released scores: the perturbation unit and the observed source/task weighting. The primary question is whether pairwise zero-exclusion decisions survive those choices; we propose no new bootstrap and claim no identified population sampling law. On 21DF the organizer-family count changes from 5/6 to 0/6, and on a single-source SpoofCeleb analysis specified before score access it changes from 6/6 to 3/6. A disclosed post-result factor decomposition localizes the latter pattern to 5/6 under speaker-only and 3/6 under attack-only perturbation. Composition policies then localize the stable conclusion: separation between the SSL-detector and organizer-baseline cohorts survives, whereas within-cohort ordering does not. Coverage stress and external families bound rather than widen that fixed-data claim.

Scope. The primary pool comprises eight author or organizer releases with high provenance, not the universe of public re-scores. We analyze all eleven Speech DF Arena re-scores [4] separately. No full-21DF "resolved" label transfers to new speakers or attacks: four source/task blocks have only 4–6 speakers each. Bands and composition radii are descriptive procedure outputs on fixed scores, not population confidence or safety claims.

2. RELATED WORK

This benchmark's own reporting. ASVspoof 2021 Fig. 4(c) pairs systems with the Bengio–Mariéthoz test [7] under Holm correction and publishes group structure [1]; Wang and Yamagishi [8] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. Poh, Martin and Bengio [9] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [10] and developed a paired correlation-corrected two-system test [11]. Bolle et al. proposed a subsets bootstrap [12], while Fogliato et al. found undercoverage for dyadic 1:1 matching [13]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [14], a delicate design [15, 16] distinct from nested metrics [17]. Speech bootstrap intervals also include utterance- and block-based forms [18, 19]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with Fig. 4(c) of the ASVspoof 2021 report nor population confidence.

Benchmarks that cannot resolve what is claimed on them. Zhuang et al. [20] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, aggregation weights change conclusions [21], channel presentation can dominate model choice [22], and other audits question representations [23] and corpus overlap [24]. None gives pairwise composition-reversal witnesses or names a dependence unit for uncertainty. Simultaneous rank intervals are established [25]; adjacency is score-selected, so we control multiplicity over the whole family.

3. METHOD

3.1. Identified object: fixed-data procedure sensitivity

Two deterministic systems on a frozen trial list have *no* sampling uncertainty; an interval exists only after declaring what was sampled. After orienting scores so larger means bona fide, we select the first score position minimizing $|\mathrm{FRR} - \mathrm{FAR}|$ and set EER to their mean without interpolation. For ordered pair (A, B), $\Delta\mathrm{EER}_{A,B} = 100(\mathrm{EER}_A - \mathrm{EER}_B)$ percentage points (negative favors A); every replicate refits the threshold. The organizers' test treats trials within class as independent draws [7]. Our original working target instead treated speakers and attacks as exchangeable with a represented evaluation design. The five spoof strata are ASVspoof, VCC2018-HUB, VCC2018-SPO, VCC2020-T1 and VCC2020-T2; the four VCC strata contain only 4/4/4/6 spoof speakers. We therefore do not replace the organizers' population claim with another unidentified one. The identified object is how paired ΔEER outputs on the fixed score set change under declared trial-i.i.d. and speaker–attack perturbation laws; confidence language is reserved for a future target with a defensible sampling law and replication.

3.2. Clustered bootstrap and simulation-checked variants

Speakers and attacks are resampled independently with replacement and trial weights multiply the two counts; bona fide trials carry no attack label and so receive speaker weights only. The primary output is the two-way percentile bootstrap. Let $\hat{\theta}$ be the eight-system vector of refitted EERs. For factor $f \in \{s, a, sa\}$ with G_f groups, delete-one vectors $\hat{\theta}_{-g,f}$ and their mean $\bar{\theta}_f$, we compute $\Sigma_f = (G_f - 1)G_f^{-1} \sum_g (\hat{\theta}_{-g,f} - \bar{\theta}_f)(\hat{\theta}_{-g,f} - \bar{\theta}_f)^\top$; sa denotes the 1,062 observed speaker$\times$attack cells [14]. The Gaussian corroborator uses the single PSD matrix $\Sigma_+ = \Sigma_s + \Sigma_a$; the same Σ_+ supplies every pair SE and its all-28-pair max-t critical value ($q = 2.878$). Because it adds the two marginal matrices without subtracting their overlap, it deliberately double-counts speaker$\times$attack cell variation. This is a conservative sensitivity construction, not an exact multiway estimator or coverage claim. Product and Gaussian Σ_+ outputs agree on all 28 zero-exclusion labels (18/28; organizer 0/6).

3.3. All-pair multiplicity for procedure bands

From $B{=}5000$ clustered replicates we form simultaneous procedure bands [26]. With $s_p = \mathrm{sd}_b(\Delta_p^{(b)})$, $q_{.95}$ is the empirical 95th percentile of $\max_p |\Delta_p^{(b)} - \bar{\Delta}_p|/s_p$ over all 28 pairs, and the reported band is $\hat{\Delta}_p \pm q_{.95}s_p$. Because the required sampling law is not identified for full 21DF, neither is called a population confidence set.

Reconstructing the cited test. Fig. 4(c) of Yamagishi et al. [1] does not disclose which Bengio–Mariéthoz variant produced its matrix, so we make no cell-level attribution. With each system's non-interpolated EER threshold selected on the evaluation set, our independent-trial and shared-trial reconstructions separate the same five of six baseline pairs. Holm is step-down, but the result is family-size independent: in the independent-trial reconstruction the weakest significant pair has $p = 9.7 \times 10^{-6} < 0.05/561$, while the sixth has $p = 0.217 > 0.05$. Because the cited derivation assumes thresholds fixed outside the test set, these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Coverage check

On the real trial index, a fitted bivariate random-effects DGP sets bona-fide scores to speaker effect plus residual, and spoof scores to offset plus speaker, attack, speaker×attack and residual effects. Its 48 cells cross organizer-like/low-EER regimes; interaction shares 0, .25, .50 and .75; Gaussian or Student-t_5 effects; and true ΔEER of 0, .5 or 2 points. All 24 organizer-like cells pass (coverage .928–.957 for delete-one inclusion–exclusion and .946–.968 for product), but at low EER the three candidate procedures have minima .845/.845/.855 and all fall below .90 together in 11/24 cells. The prespecified universal-coverage criterion is therefore rejected. This diagnoses procedure behavior under the imposed DGP, not adequacy of that DGP for 21DF.

3.5. Composition sensitivity on fixed scores

The bona-fide policies are empirical, equal-source, and equal-source/equal-speaker weighting. The spoof policies are empirical, equal-stratum, equal observed speaker×attack cell within stratum, and equal attack-family/attack/speaker-cell weighting within stratum; their Cartesian product gives 12 policies. All trial weights remain positive and class-normalized. If w_c^0 and w_c are the empirical and alternative trial-mass vectors for class c, $r_{TV} = \max_c \frac{1}{2}\sum_i |w_{c,i} - w_{c,i}^0|$. For each pair we report the policy range and empirical-to-policy paths; these are deterministic estimands, not confidence intervals. Because two initial numerical implementations failed verification, a subsequent outcome-informed search is used only to exhibit constructive upper bounds. It tests fixed Sobol and constrained differential-evolution directions, contracts each toward empirical weights, and accepts a reversal only when integer group-count, direct weighted-trial and equal-score-threshold EER computations agree. Search failure is not stability; a reported bound is not a global minimum.

4. EXPERIMENTS

Setup. The primary pool contains eight high-provenance ASVspoof 2021 DF releases: four self-supervised systems (XLSR-Mamba [27], XLS-R+SLS [28], XLSR-Conformer [29], SSL-AASIST [30]) from their authors and four organizer baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM) from the challenge key package. We use the 533,928 eval-phase trials of 611,829 released. Pooled EERs run 1.88–2.85 for the self-supervised systems and 22.38–25.56 for the baselines, reproducing published values where they exist. Pairwise intervals use B=5000 clustered replicates (seed 20260817); the 48-cell coverage stress test uses B=500 per replicate over R=1000 replicates per cell (seed 20260824).

Primary 21DF sensitivity result (Fig. 1). Both reconstructions of Fig. 4(c) separate the same five of six organizer pairs; we claim no cell-level agreement with the published matrix. A matched all-28-pair max-t bootstrap refits the same non-interpolated weighted EER in every replicate: trial-i.i.d.

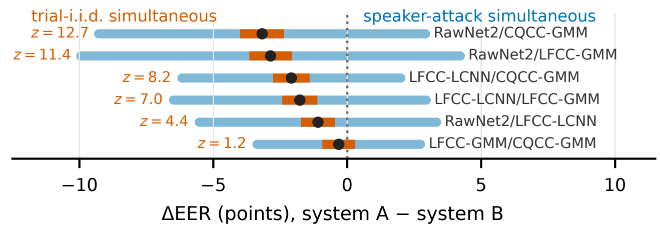

Fig. 1. The six baseline pairs, with matched all-28-pair max-t bands. Narrow bars: simultaneous trial-i.i.d. output; wide bars: simultaneous speaker–attack output. Our independent-trial reconstruction's z is shown at left and gives the same five verdicts as the narrow bars. Dots: point estimates. All six speaker–attack bands include zero.

perturbation gives 26/28 and 5/6 organizer exclusions, against 18/28 and 0/6 under speaker–attack product weights. Gaussian Σ_+ bands give the same 18/28 and 0/6. The matched threshold analysis removes threshold treatment as an explanation. A disclosed post-failure control also gives 18/28 and 0/6 while restoring each class-by-source/task mass exactly; it shows that movement in that mass is not necessary for the contrast, but is a constructive robustness control rather than a prospective causal decomposition. The widest 3.18-point gap has product band $[-9.27, 2.91]$ and Gaussian Σ_+ band $[-8.76, 2.40]$.

Single-source test specified before score access. Before inspecting SpoofCeleb scores, we fixed the detector roster, score processing, perturbations, seed and six-pair max-t endpoint. On the complete 91,130-trial grid [31], four fixed off-domain detectors give 6/6 trial-i.i.d. versus 3/6 product exclusions. AASIST, SLS, SSL-AASIST and XLSR-Mamba have point EERs 57.93%, 24.51%, 26.72% and 27.58%, respectively; this check concerns a weakly transferred off-domain family, not high-performing in-domain SpoofCeleb systems. Every base item has one bona-fide and nine spoof renditions under a single source label. Source composition is therefore fixed by design, although product draws still vary speaker and spoof-attack multiplicities; between-source movement is not necessary for the contrast. A disclosed post-result decomposition gives 5/6 speaker-only and 3/6 attack-only exclusions, the latter reproducing the joint verdict vector: under these procedures attack-level perturbation alone is sufficient for all three changes, while speaker-only is sufficient for one. A later reproducibility audit found that the archived specification omitted the scoring executable. A disclosed post-result rerun from clean, identified source revisions reproduced three score files byte-for-byte; XLSR-Mamba differed by at most 1.44×10^{-6}, with all EERs, decisions and the prespecified 6/6-to-3/6 result unchanged.

What the benchmark supports (Table 1). All six organizer pairs straddle zero under product and Gaussian Σ_+ perturbations despite a 3.18-point span. RawNet2–CQCC-

Table 1. Within-cohort pairs: empirical $|\Delta\text{EER}|$ and product max-t simultaneous half-width (points), plus the smallest verified constructive r_{TV} upper bound. Conf=Conformer; SSL-AAS=SSL-AASIST. A bound is a found reversal, not a minimum or confidence limit.

pair A	B	$\|\Delta\text{EER}\|$	sim. half-w.	$r_{TV} \leq$
XLSR-Mamba	XLS-R+SLS	0.032	0.693	0.011
XLSR-Mamba	XLSR-Conf	0.389	0.959	0.344
XLSR-Mamba	SSL-AAS	0.968	0.860	0.246
XLS-R+SLS	XLSR-Conf	0.357	1.136	0.172
XLS-R+SLS	SSL-AAS	0.935	0.848	0.434
XLSR-Conf	SSL-AAS	0.578	1.059	0.166
RawNet2	LFCC-LCNN	1.094	4.434	0.069
RawNet2	LFCC-GMM	2.864	7.085	0.094
RawNet2	CQCC-GMM	3.180	6.104	0.112
LFCC-LCNN	LFCC-GMM	1.770	4.720	0.087
LFCC-LCNN	CQCC-GMM	2.086	4.087	0.104
LFCC-GMM	CQCC-GMM	0.316	3.054	0.034

GMM, LFCC-LCNN–LFCC-GMM and LFCC-LCNN–CQCC-GMM become zero-excluding in leave-one-corpus-out refits, so the organizer result is not composition-robust. The 16 between-cohort gaps are 19.5–23.7 points and survive every deletion. Two within-SSL zero-exclusions at 0.97 and 0.94 points are withdrawn because low-EER stress is anti-conservative and both disappear after source deletion; adjacent within-SSL margins are only 0.03, 0.36 and 0.58 points.

Composition robustness. Under the 12 registered score-blind policies, 6/12 within-cohort orderings reverse and 0/16 between-cohort orderings do; one of the 11 prespecified empirical-to-policy paths reverses before $r_{TV} = .10$. The later outcome-informed search cannot revise that primary result: it verifies 729 direction endpoints and supplies constructive upper bounds for all 12 within-cohort reversals, five at $r_{TV} \leq .10$. XLSR-Mamba versus XLS-R+SLS reverses at .010218. All three EER computations agree, but the delta at that witness is tiny; the result is local sensitivity, not practical superiority of either system. No between-cohort witness was found, which is a search outcome rather than proof of global stability.

Independent re-score layer. Treating the eleven complete Speech DF Arena re-scores as a separate 55-pair provenance stratum changes the zero-exclusion count from 52 under trial-i.i.d. perturbation to 38 under speaker–attack perturbation; the median pointwise-width ratio is 8.45. These are descriptive sensitivity outputs, not formal Arena confidence bands.

Speakers are not exchangeable either. The bona-fide sources ASVspoof, VCC2018 and VCC2020 contain 67, 12 and 14 speakers, respectively. Leave-one-source-out delete-one refits preserve all 16 between-cohort verdicts, remove both within-SSL resolutions and add three baseline ones. Thus cohort separation survives these provenance changes, while every within-cohort resolution is conditional on them.

External family-level check. On the ASVspoof 5 Track 1 evaluation set [3], EER bands for SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba over all 680,774 trials meet both prespecified criteria: at least one trial-i.i.d. exclusion disappears when the 367 target speakers and 370 bona-fide-only non-target speakers are resampled separately and the 16 spoof attacks independently, and median width inflation is $27.27\times$. This is an external fixed-family sensitivity check, not pair-level replication or population inference.

Measured width inflation. On the real scores, simultaneous product bands are 4.13–$11.36\times$ wider than matched trial-i.i.d. bands over all 28 pairs (median $8.14\times$). Under the fitted organizer-like Gaussian DGP on the real incidence, a witnessed R=200 run gives 18.5% trial-i.i.d. coverage (Wilson 13.7–24.5%) versus 97.5–98.0% for the clustered candidates (combined Wilson endpoints 94.3–99.2%). This evaluates repeated-sampling coverage under the imposed DGP; it does not validate that DGP as a model of 21DF.

5. DISCUSSION

For reporting, and limits. Plans, deviations, hashes and receipts are at `github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty`. Legacy SSL-AASIST/AASIST NPZs lack historical run provenance, and the historical freeze lacks an independently verifiable timestamp. Report speaker and attack incidence plus composition, not trial counts: bands require a sampling law and coverage; radii require a policy family and witness. We withdraw full-21DF population labels, modern rank-confidence claims and formal Arena bands. Zero-straddling is sensitivity, not equality.

6. CONCLUSION

On fixed released scores, matched 21DF perturbations change organizer-family exclusions from 5/6 to 0/6, and a single-source SpoofCeleb analysis specified before score access changes them from 6/6 to 3/6. Because SpoofCeleb has one source, between-source movement cannot explain the latter contrast; a post-result decomposition shows that attack-only perturbation reproduces its 3/6 verdict vector. Registered composition policies reverse 6/12 within-cohort orderings but 0/16 between-cohort orderings. Across the registered policies, corpus deletions and searched directions, the stable conclusion is separation between the four SSL detectors and four organizer baselines, not within-cohort ranking. Population confidence requires independently sampled units and a defensible acquisition law.

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